Autonomous Predator-Prey System: A Cybernetic Approach

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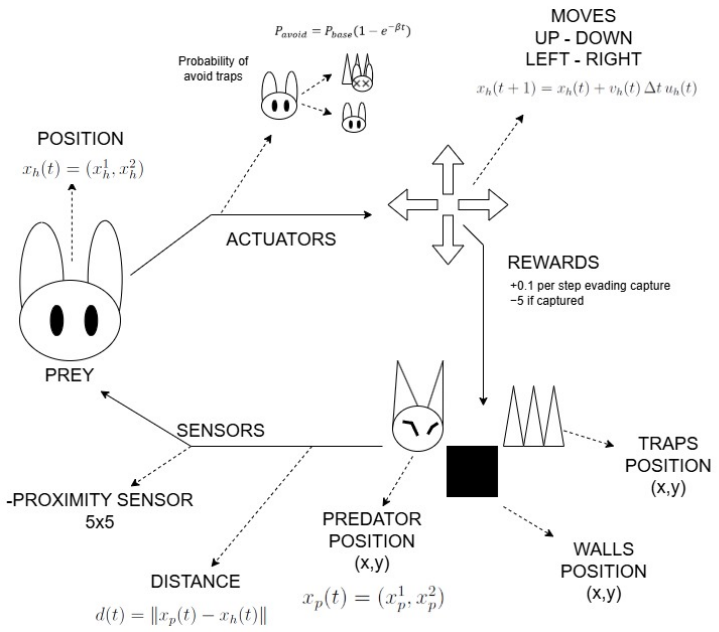
Adaptive Systems

Predator-Prey System Dynamics

Basic Ecological Model:

- Single predator and single prey
- Discrete 2D environment with obstacles
- Asymmetric objectives:
  - Predator: Maximize capture rate
  - Prey: Minimize capture probability



Basic interaction scheme

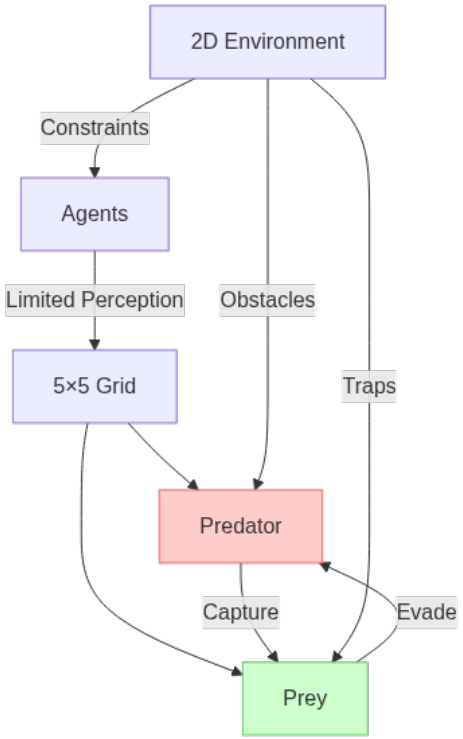
Key Characteristics

- Partial observability (5×5 grid perception)
- Continuous adaptation requirement
- Dynamic equilibrium target

Core Scientific Challenge

Dynamic System Characteristics:

- Discrete 2D environment with obstacles
- Partial observability (5×5 grid perception)
- Competitive asymmetric objectives
- Continuous adaptation requirement



System interaction dynamics

Research Question

How can cybernetic principles and machine learning be combined to create stable, adaptive predator-prey dynamics in a constrained environment?

Core Equations

Agent Dynamics

$$\pi_i(t+1) = \pi_i(t) + \Delta t \cdot r_i(t) \cdot v_i(t)$$

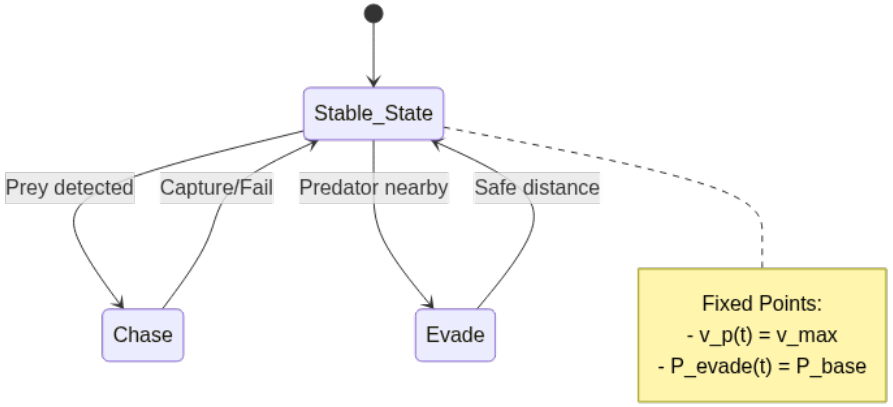
where $i \in \{p, h\}$ (predator/prey), r_i is direction, v_i is velocity

Predator Speed Adaptation:

$$v_p(t) = \nu_{\text{max}} + \frac{\alpha}{1 + e^{-kd(t)}}$$

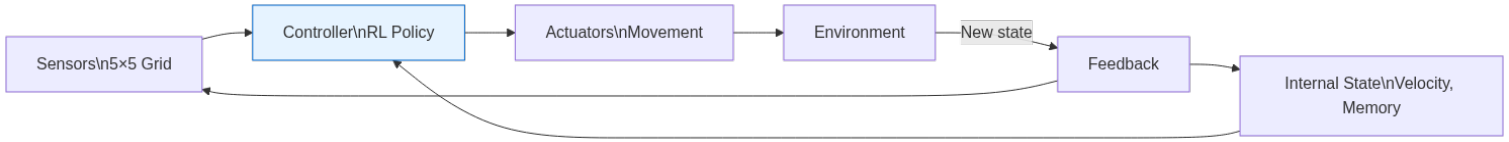
Prey Evasion Learning:

$$P_{\text{evade}}(t) = P_{\text{base}}(1 - e^{-\beta t})$$



Phase space of system dynamics

Cybernetic Feedback Structure



Components:

- Sensory perception (5×5 grid)
- Policy controller (RL/Cybernetics)
- Adaptive actuators

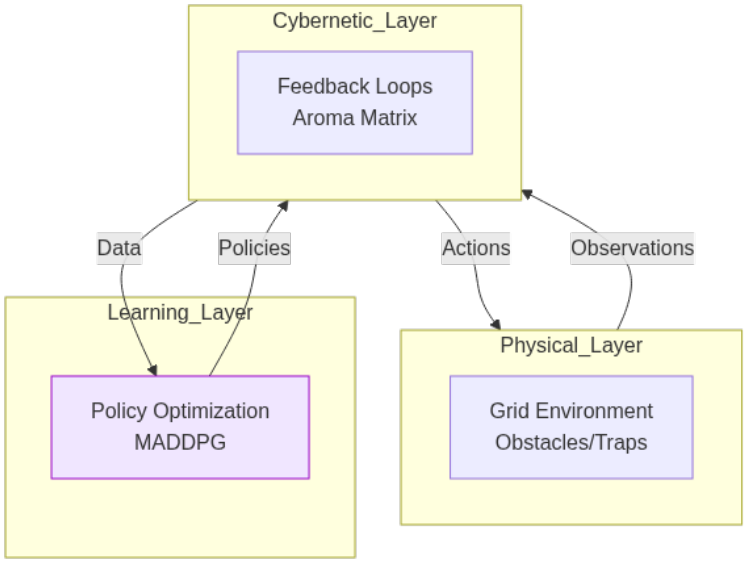
Novel Elements:

- Aroma matrix tracking
- Speed modulation
- Evasion probability

Integrated Learning-Control Framework

Three-Layer Architecture:

1. **Physical Layer**: Grid environment with obstacles
2. **Cybernetic Layer**: Feedback control loops
3. **Learning Layer**: Policy optimization



Key Innovation

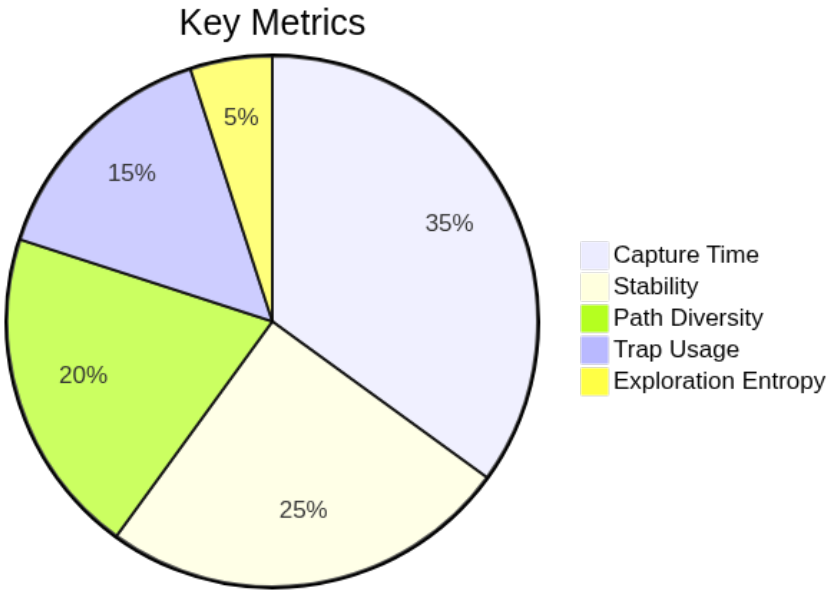
Synergy between:

- RL for strategic adaptation
- Cybernetics for real-time stability
- Dynamical systems theory for analysis

Analysis Framework

Quantitative Metrics:

- Capture time distribution
- Learning convergence
- Phase space coverage
- Entropy measures



Stability Analysis

$$\max_{\pi_p} \min_{\pi_h} \mathbb{E}[R_p - R_h] \leq \epsilon$$

Bounded competitive equilibrium

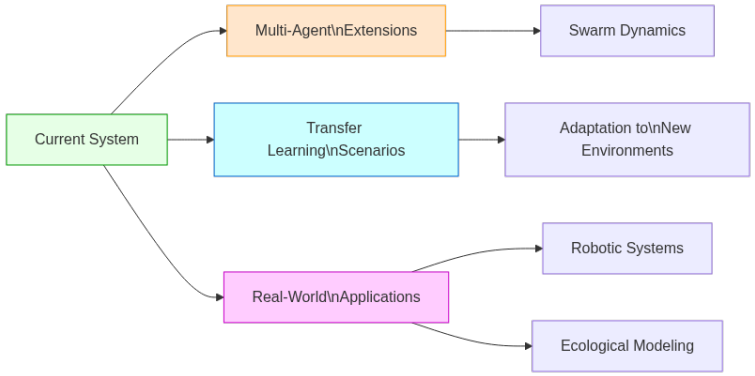
Contributions and Future Work

Key Contributions

- Novel integration of cybernetics and RL
- Analytically tractable dynamic model
- Experimentally validated emergent behaviors

Future Directions:

- Multi-agent extensions
- Transfer learning scenarios
- Real-world applications



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Thank you! Questions?